

Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly known as RAG, is a technique that combines a large language model with an external knowledge source. Instead of relying only on what the model learned during training, RAG allows the model to look up relevant information at the moment a question is asked, and then use that information to produce a more accurate and grounded answer.

A typical RAG pipeline has three main stages. The first stage is ingestion, where documents such as PDFs, web pages, or text files are loaded and split into smaller chunks. Splitting is necessary because language models can only process a limited amount of text at once, and smaller chunks make it easier to find the exact piece of information relevant to a question.

The second stage is embedding and storage. Each chunk of text is converted into a vector, which is a list of numbers that captures the meaning of the text. These vectors are stored in a vector database, such as Chroma or FAISS, which allows for fast similarity search later on.

The third stage is retrieval and generation. When a user asks a question, the question is also converted into a vector. The system then searches the vector database for the chunks whose vectors are most similar to the question. These retrieved chunks are passed to the language model along with the original question, and the model generates a final answer grounded in that retrieved context.

RAG systems are widely used in customer support assistants, internal company knowledge bases, and research tools, because they reduce hallucination and allow the model to answer questions about information it was never directly trained on, such as private documents or very recent events.

Common tools used to build RAG systems include LangChain and LlamaIndex for orchestration, OpenAI or open-source models for embeddings and generation, and vector databases such as Chroma, FAISS, Pinecone, or Weaviate for storage and retrieval. Web frameworks like FastAPI or Django are often used to expose the RAG pipeline as an API that a frontend application can call.